

IPL Match & Performance Analysis: A Data-Driven Study

Executive Summary

The Indian Premier League (IPL) is one of the most competitive and data-rich T20 cricket tournaments in the world. This project analyzes IPL match-level and ball-by-ball datasets to uncover patterns related to batting performance, bowling efficiency, toss decisions, venue influence, and match outcomes.

The datasets were cleaned, integrated, and transformed using Pandas and NumPy. Statistical analysis and visualizations were then performed using Matplotlib and Seaborn. The analysis revealed that batting performance has improved significantly over time, teams often gain an advantage while chasing, and death-over performance plays a crucial role in determining match outcomes.

The study demonstrates how data analytics can be used to identify trends, evaluate player performance, and support strategic decision-making in professional cricket.

Methodology

The project was completed in five major stages:

1. Data Collection and Understanding

Two datasets were used:

- **matches.csv** – Match-level information including season, venue, toss decisions, winners, and player-of-the-match awards.
- **deliveries.csv** – Ball-by-ball information including batter, bowler, runs scored, extras, wickets, and dismissal details.

2. Data Cleaning

The datasets were examined for missing values and inconsistencies.

The following preprocessing steps were performed:

- Missing city values were replaced with "Unknown".
- Missing player_of_match values were replaced with "No Award".
- Missing winner values were labeled as "No Result".

- Missing dismissal_kind values were replaced with "not out".
- Duplicate records were checked and removed if necessary.
- Date columns were converted into datetime format.

3. Data Integration

The match-level and delivery-level datasets were merged using the match identifier, creating a unified analytical dataset.

4. Feature Engineering

Several new variables were created:

- Total Runs Scored
- Boundary Classification
- Over Phase Classification
- Year
- Month

5. Statistical Analysis and Visualization

Various aggregation techniques and visualizations were used to identify important IPL trends and performance patterns.

Key Findings

1. Virat Kohli is the Highest Run Scorer

The batting analysis revealed that Virat Kohli is the highest run scorer across all IPL seasons, accumulating more than 8,000 runs.

Other leading run scorers include:

- Shikhar Dhawan
- Rohit Sharma
- David Warner
- Suresh Raina

This highlights their consistency and long-term contribution to their respective teams.

2. Economical Bowling Remains a Key Success Factor

Economy rate analysis was conducted for bowlers who delivered at least 500 balls.

Among the most economical bowlers were:

- Anil Kumble
- Muttiah Muralitharan
- Sunil Narine
- Dale Steyn
- Daniel Vettori

These bowlers consistently restricted scoring opportunities and maintained pressure on opposition batters.

3. Toss Decisions Influence Match Outcomes

The toss analysis indicated that teams choosing to field first often achieve a higher win percentage compared to teams batting first.

This trend suggests that chasing targets provides tactical advantages such as:

- Better assessment of match conditions
 - Improved run-rate management
 - Greater flexibility in batting strategy
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4. IPL Has Become More Batting Friendly

Season-wise scoring analysis revealed a gradual increase in average runs per match over time.

Several factors may contribute to this trend:

- More aggressive batting approaches
- Improved fitness and power hitting
- Better batting conditions
- Evolution of T20 cricket strategies

The regression trend line further confirmed a positive relationship between season progression and scoring rates.

5. Death Overs Significantly Impact Match Results

The analysis of overs 16–20 demonstrated that teams capable of scoring consistently during death overs often gain a competitive advantage.

Teams with strong death-over performance typically:

- Accelerate scoring rapidly
- Convert good totals into winning totals
- Successfully chase difficult targets

This phase remains one of the most influential periods in T20 cricket.

6. Venue Influence and Home Advantage

Venue-based analysis suggested that certain stadiums are associated with higher win frequencies for specific teams.

Possible reasons include:

- Familiar playing conditions
- Crowd support
- Pitch behavior
- Strategic team selection

These findings indicate the presence of a measurable home-ground advantage in IPL matches.

Strategic Recommendation

Based on the findings, IPL teams should prioritize two key strategies:

1. Prefer Chasing When Conditions Permit

The toss analysis indicates that teams often perform better while chasing. Captains should consider fielding first whenever pitch and weather conditions remain stable.

2. Invest in Death-Over Specialists

The final overs contribute significantly to overall match outcomes. Teams should focus on:

- Recruiting powerful finishers
- Developing specialist death-over bowlers
- Practicing high-pressure end-game scenarios

These improvements can directly influence winning percentages.

Limitations

Although the dataset provided valuable insights, several important factors were unavailable:

- Weather conditions
- Pitch reports
- Player fitness information
- Injury records
- Team combinations
- Advanced player tracking data

Including these variables could enable deeper analysis and more accurate predictive modeling.

Conclusion

This project successfully applied data science techniques to analyze IPL match and delivery data. Through data cleaning, feature engineering, aggregation, and visualization, several meaningful patterns were discovered.

The analysis demonstrated the dominance of elite batters, the value of economical bowling, the importance of toss decisions, and the growing significance of death-over performance. Furthermore, venue-based trends highlighted the potential impact of home advantage.

Overall, the project shows how data-driven approaches can provide actionable insights for teams, coaches, analysts, and cricket enthusiasts. The findings reinforce the growing role of analytics in modern cricket strategy and decision-making.